

cp_2d

1.

Between 25 and 75 | 🟡 20/20

Write a program that reads a number, and checks if the given number is between **25** and **75**.

Input

The input will be a single line containing an integer.

Output

The output should be a single line containing a boolean. **True** should be printed if the given number is between **25** and **75**, otherwise **False** should be printed.

Explanation

For example, if the given number is **35**,

- The number **35** is greater than **25**.
- The number **35** is less than **75**.
- So, the number **35** is between **25** and **75**.

The output should be **True**.

Sample Input 1

35

Sample Output 1

True

Sample Input 2

20

Sample Output 2

False

Sample Input 3

75

Sample Output 3

False

code:

```
n=int(input())
print(n>25 and n<75)
```

2.

Women | 🟡 15/15

Easy

Write a program that reads three strings `A` , `B` , and `C` , and checks if one of the given strings is equal to `woman`.

Input

The first line of input contains a string `A` .
The second line of input contains a string `B` .
The third line of input contains a string `C` .

Output

The output should be a single line containing a boolean. `True` should be printed if one of the given strings is equal to `woman`, otherwise `False` should be printed.

Explanation

For example, if the given strings are `A = "man"` , `B = "woman"` , and `C = "man"` ,

X `A` is equal to `woman`. (`man` is not equal to `woman`)

✓ `B` is equal to `woman`. (`woman` is equal to `woman`)

X `C` is equal to `woman`. (`man` is not equal to `woman`)

The output should be `True` as one of the given strings is equal to `woman`.

Sample Input 1

```
man
woman
man
```

Sample Output 1

```
True
```

Sample Input 2

```
man
man
man
```

Sample Output 2

```
False
```

code:

```
n=input()
m=input()
o=input()
p=(n=="woman" or m=="woman" or o=="woman")
print(p)
```

3.

Less than 15 | 🟡 15/15

Easy

Write a program that reads three numbers `A` , `B` , and `C` , and checks if any of the given numbers is less than `15`.

Input

The first line of input contains an integer `A` .
The second line of input contains an integer `B` .
The third line of input contains an integer `C` .

Output

The output should be a single line containing a boolean. **True** should be printed if any of the given numbers is less than `15`, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 18` , `B = 2` , and `C = 20` ,

X `A` is less than `15`. (`18` is not less than `15`)

✓ `B` is less than `15`. (`2` is less than `15`)

X `C` is less than `15`. (`20` is not less than `15`)

The output should be **True** as one of the given numbers is less than `15`.

Sample Input 1

```
18
2
20
```

Sample Output 1

```
True
```

Sample Input 2

```
65
25
451
```

Sample Output 2

```
False
```

code:

```
A=int(input())
B=int(input())
C=int(input())
d=(A<15 or B<15 or C<15 )
print(d)
```

4.

Not Negative | 🟡 20/20

Easy

Write a program that reads two numbers `A` and `B` , and checks if both the given numbers are **not negative**.

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .

Output

The output should be a single line containing a boolean. **True** should be printed if both the given numbers are **not negative**, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 5` and `B = 10` .

✓ **A** is not a negative number. (**5** is **not** a negative number)

✓ **B** is not a negative number. (**10** is **not** a negative number)

The output should be **True** as both the given numbers are **not negative**.

Sample Input 1

```
5
10
```

Sample Output 1

```
True
```

Sample Input 2

```
-30
421
```

Sample Output 2

```
False
```

code:

```
A=int(input())
B=int(input())
print(A>0 and B>0)
```

5.

Not Less than 15 | 🟡 20/20

● Solved

Easy

Write a program that reads two numbers `A` and `B`, and checks if any of the given numbers is **not less than 15**.

Input

The first line of input contains an integer representing `A`.
The second line of input contains an integer representing `B`.

Output

The output should be a single line containing a boolean. **True** should be printed if any of the given numbers is **not less than 15**, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 921` and `B = 6`,

✓ **A** is not less than 15. (921 is **not less than 15**)

✗ **B** is not less than 15. (6 is less than 15)

The output should be **True** as one of the given numbers is **not less than 15**.

Sample Input 1

```
921
6
```

Sample Output 1

```
True
```

Sample Input 2

```
1
10
```

Sample Output 2

```
False
```

code:

```
A=int(input())
B=int(input())
print(A>=15 or B>=15)
```

6.

Easy

Write a program that reads an age `A` and guardian status `S`, and checks if the age `A` is between **12** and **60** or if the guardian status `S` is equal to **yes**.

Note

The guardian status will be either **yes** or **no**

Input

The first line of input contains an integer representing the age `A`.

The second line of input contains a string representing the guardian status `S`.

Output

The output should be a single line containing a boolean. **True** should be printed if the age is between **12** and **60** or if the guardian status `S` is equal to **yes**, otherwise **False** should be printed.

Explanation

For example, if the given age `A = 15` and guardian status `S = "no"`,

✓ `A` is between **12** and **60**. (15 is between **12** and **60**)

X `S` is equal to **yes**.

The output should be **True** as the age is between **12** and **60**.

Sample Input 1

```
15
no
```

Sample Output 1

```
True
```

Sample Input 2

```
61
no
```

Sample Output 2

```
False
```

code:

```
age=int(input())
gaurdian_status=input()
print(((age>12)and (age<60)) or gaurdian_status=="yes")
```

7.

Compare Numbers - 5 | 🏆 20/20

● Sol

Easy

Write a program that reads two numbers `A` and `B` and checks if one of the given numbers is negative and product of the numbers is greater than or equal to `-46`.

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .

Output

The output should be a single line containing a boolean. **True** should be printed if one of the given numbers is negative and product of the numbers is greater than or equal to `-46`, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = -10` and `B = 1` ,

✓ One of the given numbers is negative. (`-10` is negative)

✓ Product of the numbers is greater than or equal to `-46`. (`-10 * 1 = -10` , `-10` is greater than or equal to `-46`)

The output should be **True** as one of the given numbers is negative and product of the numbers is greater than or equal to `-46`.

Sample Input 1

```
-10
1
```

Sample Output 1

```
True
```

Sample Input 2

```
5
2
```

Sample Output 2

```
False
```

code:

```
first_number=int(input())
second_number=int(input())
d=first_number<0 or second_number<0
d1=(first_number*second_number)>=-46
print(d and d1)
```

8.

Greater than Five | 🟡 20/20

● S

Write a program to check if both of the given numbers are positive and if atleast one of them is greater than 5.

Note

Zero is not a positive number.

Input

First line of the input will be an integer
Second line of the input will be an integer

Output

Print `True` if both the numbers are positive and if atleast one of the numbers is greater than 5.
In all other cases print `False`

Explanation

When the given numbers are -10 and 6:

- ✓ At least one number is greater than 5 (6 is greater than 5)
- ✗ Both the numbers are positive (-10 is a negative number.)

When the given numbers are 10 and 1:

- ✓ At least one number is greater than 5 (10 is greater than 5),
- ✓ Both the numbers are positive.

Sample Input 1

```
-10
6
```

Sample Output 1

```
False
```

Sample Input 2

```
10
1
```

Sample Output 2

```
True
```

code:

```
num1=int(input())
num2=int(input())
d=(num1>0 and num2>0)
d1=(num2>5 or num1>5)
print(d1 and d)
```

9.

Medium

Write a program that reads the marks in Maths M , Physics P , and Chemistry C , and checks if any of the below conditions is satisfied.

- $M \geq 70$ and $P \geq 60$ and $C \geq 60$
- $M + P + C \geq 180$

Input

The first line of input contains an integer representing M .
 The second line of input contains an integer representing P .
 The third line of input contains an integer representing C .

Output

The output should be a single line containing a boolean. **True** should be printed if any of the given conditions is satisfied, otherwise **False** should be printed.

Explanation

For example, if the given marks are $M = 0$, $P = 100$, and $C = 100$,

X $M \geq 70$ and $P \geq 60$ and $C \geq 60$ (0 is less than 70)

✓ $M + P + C \geq 180$. ($0 + 100 + 100 = 200$, 200 is greater than 180)

The output should be **True** as one of the given conditions is satisfied.

Sample Input 1

```
0
100
100
```

Sample Output 1

```
True
```

Sample Input 2

```
30
40
65
```

Sample Output 2

```
False
```

code:

```
maths=int(input())
phy=int(input())
chem=int(input())
val=(maths>=70 or phy>=60 or chem>=60)
s=(maths+phy+chem)>=180
print(val and s)
```

10.

Compare with Ten | 🟡 25/25

● Solvec

Easy

Write a program that reads two numbers `A` and `B` and check if any one of the below conditions is satisfied.

- The sum of `A` and `B` is less than **10**.
- The difference between `A` and `B` is less than **10**.
- `A` is between **5** and **30**.

Input

The first line of input contains an integer representing `A`.

The second line of input contains an integer representing `B`.

Output

The output should be a single line containing a boolean. **True** should be printed if any one of the conditions is satisfied, otherwise **False** should be printed.

Explanation

For example, if the given numbers are `A = 12` and `B = 8`,

✗ The sum of `A` and `B` is less than **10**. ($12 + 8 = 20$, **20** is not less than **10**)

✓ The difference between `A` and `B` is less than **10**. ($12 - 8 = 4$, **4** is less than **10**)

✓ `A` is between **5** and **30**. (**12** is between **5** and **30**)

The output should be **True** as one of the conditions is satisfied.

Sample Input 1

```
12
8
```

Sample Output 1

```
True
```

Sample Input 2

```
50
15
```

Sample Output 2

```
False
```

code:

```
n1=int(input())
n2=int(input())
c1=(n1+n2)<10
c2=(n1-n2)<10
c3=(n1>5) and (n1<30)
print(c1 or c2 or c3)
```